

Figure 1: Tempest

DevStack

DevStack is an opinionated script to help you quickly create an OpenStack development environment. It can also be used to demonstrate how to start and run OpenStack services, and provides examples of using them from a command line.

DevStack, as the name itself suggests, is used specifically for development work on OpenStack. The OpenStack/DevStack environment is set up after running `stack.sh` (a script to start DevStack configuration). From Git repositories we can download the latest version of DevStack.

DevStack has evolved to support many configuration options, alternative platforms and support services. In this article, I have considered Ubuntu 14.04, CentOS 7.0 and Fedora 21. More on DevStack and its configurations will be described later.

Some well-known features of OpenStack/DevStack are: Neutron, Cinder, Keystone, Glance, Swift, Nova and Horizon. Not all the features are covered and explained here.

The basic Tempest reference architecture

Tempest is an OpenStack integration test suite based on the unittest2 framework and currently uses the Nosetest runner.

Tempest is a set of integration tests to be run against any live OpenStack/DevStack cluster. It has a collection of tests for OpenStack API validation, different scenarios and other specific tests that are useful in validating OpenStack deployment. Tempest also drives some load in the OpenStack cloud environment, so that some part of load testing is also

Basic Tempest Reference Architecture

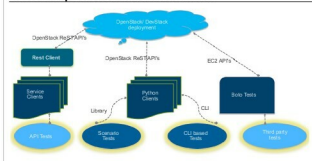


Figure 2: Detailed Tempest reference architecture

Basic Tempest Reference Architecture

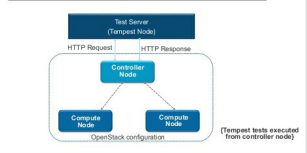


Figure 3: Basic Tempest reference architecture

covered. This is possible because it runs all API validation tests in parallel.

Overall, all Tempest does is run tests against OpenStack service endpoints by exercising REST API calls and validate the responses.

Tempest is not exactly tied to any single test runner, but 'testr' is the most commonly used tool. Once all the required configuration is done, Tempest can be run in various ways:

- 1) Tests can be run in parallel.
- 2) Tests can be run by selecting individual components serially.
- 3) Using the 'tox' command, tests can be run again in two ways.
 - `tox -efull`: This will run the same set of tests as the OpenStack gate.
 - `tox -esmoke`: This is to run the tests tagged as 'smoke'.
- 4) Tempest also has a set of unit tests that test the Tempest code itself. These tests can be run by the `run_tempest.sh` command.

The various types of tests that can be executed using Tempest are:

- Smoke tests
- Positive tests
- Negative tests
- Stress tests
- White box tests

Tempest is based on `httplib2` and built using the REST framework.

The bare minimum hardware required for OpenStack/

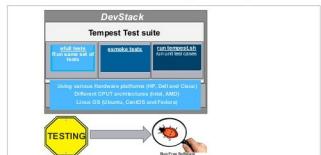


Figure 4: Tempest test suite